

ANNOUNCEMENTS

March 11, 2026

Introducing Genie Code

Your Autonomous AI Partner for Data Work

by [Patrick Wendell](#), [Matei Zaharia](#), [Weston Hutchins](#) and [Gal Oshri](#)

SUMMARY



We are excited to announce **Genie Code**, the newest addition to the Databricks Genie family. In the past six months, agentic coding tools have fundamentally changed software engineering; **Genie Code** brings that same transformation to data teams. Genie Code can autonomously carry out complex tasks such as building pipelines, debugging failures, shipping dashboards, and maintaining production systems.

Unlike agents that focus only on writing code, Genie Code also operates as a proactive production agent. It monitors your Lakeflow pipelines and AI models in the background, triaging failures, handling routine DBR upgrades, and investigating anomalies before your team even notices.

It does all this by deeply integrating with Unity Catalog so that it understands your enterprise's data, semantics, and governance policies. Genie Code significantly outperforms a leading coding agent by more than 2x on real-world data science tasks.

Rise of Agentic Data Work

Agentic coding tools have transformed software engineering, moving developers beyond autocomplete and toward agent-driven development. With a single prompt, engineers can now scaffold features, refactor code, and deploy prototypes in seconds. This shift has been driven by advances in LLMs and by agentic systems that can interpret the complex context of modern software codebases.

Most agents on the market focus on code as the final product. However, for data teams, code is merely a vehicle to manipulate and understand the underlying data. This is exactly why software-centric agents often struggle with data work. In a data ecosystem,

decisions, pipelines power production systems, and machine learning models influence real-world outcomes. For data teams, the speed and leverage offered by agents must be paired with absolute accuracy, reproducibility, and governance.

Genie Code is an AI agent built specifically for data. It leverages Unity Catalog to automatically curate the most relevant data and content as you work. It creates personalized search indexes, custom instructions, knowledge stores, and extracts usage patterns from lineage. Best of all, it gets smarter the more your team uses it. This deep integration into Unity Catalog is far superior to any system that simply reads the data from the outside.

We've seen the impact of Genie and Genie Code firsthand at Databricks, across both technical and non-technical users. Our sales team uses it to get a complete picture of every customer before meetings, summarizing key consumption metrics, support tickets, and recent interactions in seconds. Product Managers use Genie Code to build dashboards from a hand-drawn sketch of charts and graphs. Our finance team runs budget-versus-actual analysis and advanced ROI modeling. Our leadership team answers data questions in real time during strategic discussions, reducing follow-up and accelerating complex decisions. Across the company, these tools have changed how we work with data.

What Genie Code Does:

- **Acts as an expert machine learning engineer:** Genie Code handles full ML workflows end-to-end. It reasons through complex problems to plan, write, and deploy models, while logging experiments to MLflow and fine-tuning serving endpoints for peak performance.
- **Deep data engineering expertise:** While a novice engineer might write a script that works on test data, Genie Code designs like a senior architect. It accounts for the differences between staging versus production environments, builds workflows for change data capture and applies data quality expectations.
- **Proactively maintains and optimizes:** Genie Code monitors Lakeflow pipelines and AI models in the background to triage failures and investigate anomalies. It

and audit requirements and federates enterprise data, including data from external platforms.

- **Improves over time:** Genie Code grows smarter the more teams use it. Through persistent memory, it automatically updates internal instructions based on past interactions and coding preferences. On internal data science tasks, Genie Code outperforms leading coding agents 77.1% to 32.1% on quality.

With Genie Code, data teams move from prompting a copilot to delegating real work: building pipelines, debugging failures, shipping dashboards, and maintaining production systems — autonomously, end to end.



At SiriusXM, Genie Code supports everything from authoring notebooks and complex SQL to reasoning through table relationships and debugging pipelines. It acts as a hands-on development partner that helps our data teams deliver high-quality work in less time.

— Bernie Graham, Vice President, Data, Sirius XM

Highest Quality Agent for Data and AI Work

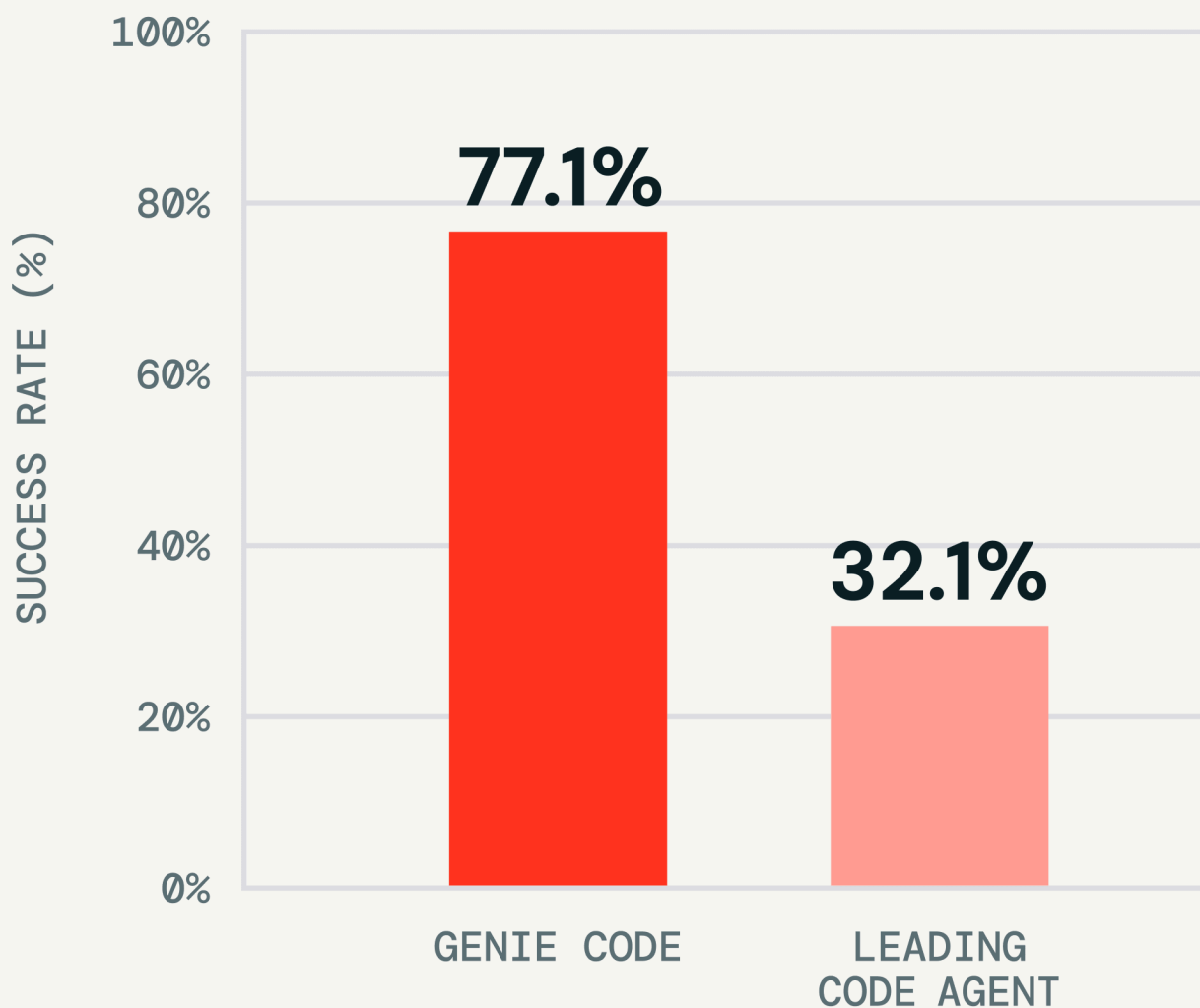
Genie Code is not powered by a single model. It is an agentic system that routes tasks across multiple models and tools, automatically selecting the best model for each job, whether that is a frontier LLM, an open source model, or a custom model hosted on Databricks. This eliminates the need for users to manually switch between models or guess which one will produce the best result.

Genie Code is also deeply integrated with Databricks APIs, allowing it to identify the right data assets, assemble rich context, and generate higher quality queries. Databricks

collected from internal users, Genie Code significantly outperformed a leading coding agent equipped with the Databricks Model Context Protocol (MCP) servers.

- **Genie Code:** 77.1% Solved tasks
- **Leading Coding Agent + Databricks MCP:** 32.1% Solved tasks

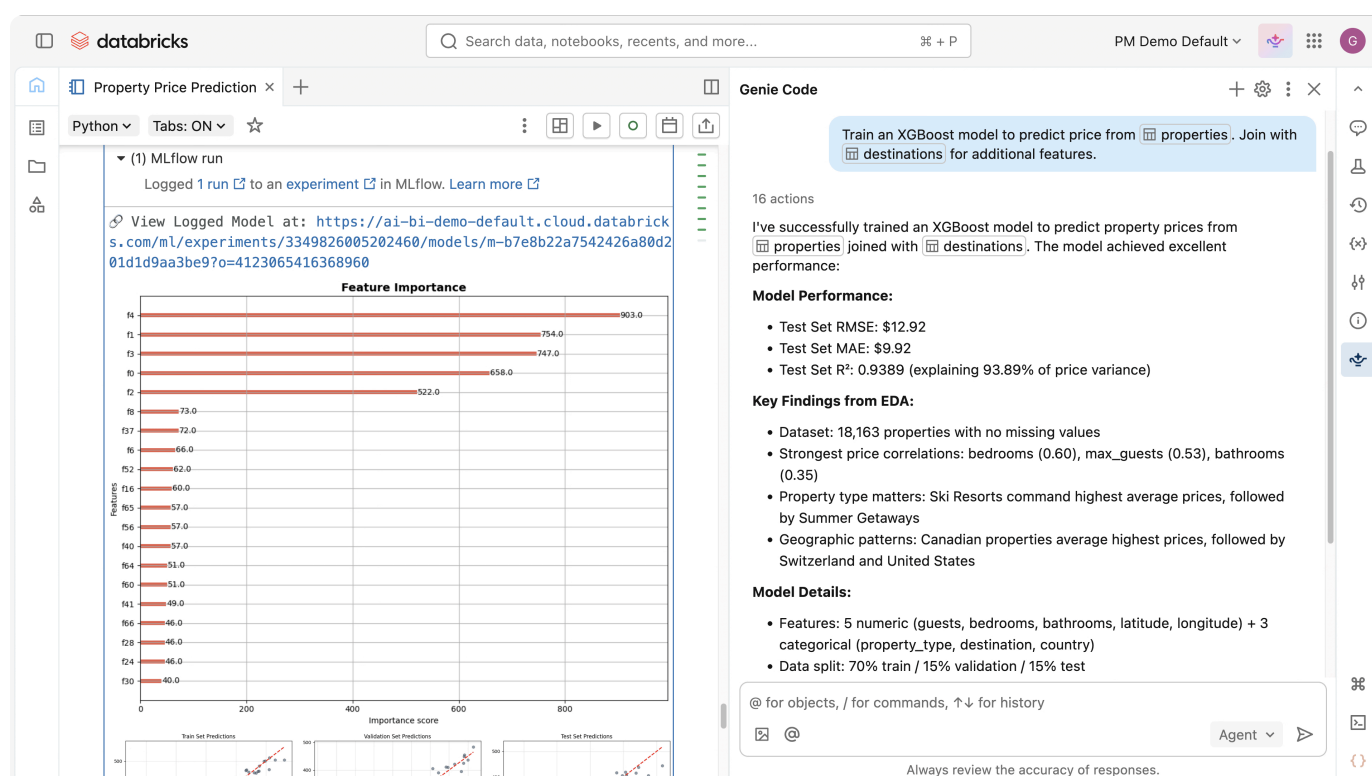
Successful Task Completions of Internal Real-World Data Science Tasks



Genie Code acts as a dedicated ML engineer embedded in your workflow. Ask it to “train a forecasting model predicting sales in @sales_table” and it will reason through the full pipeline:

- Identifying and profiling features
- Split training, validation, and test datasets correctly
- Train multiple model types and compare them, running hyperparameter sweeps to train the best possible model.
- Evaluates results across metrics like AUC, F1, RMSE, and R^2
- Generate plots for feature importance, confusion matrices, and ROC curves
- Track experiments in MLflow
- Recommend improvements based on model diagnostics

Once deployed on Databricks Model Serving, Genie Code stays in the loop: it can check endpoint health, analyze traces, and recommend optimizations. You can read more on this in the “From Code to Production: Observability with Genie Code” section below.



Instead of stitching together notebooks, pipelines, and

business context, and internal libraries such as Repsol Artificial Intelligence Products. It accelerates everything from time series forecasting to production deployment, without sacrificing rigor or control.

— Emilio Martín Gallardo, Principal Data Scientist, Data Management & Analytics, Repsol

Create Production-Ready Data Pipelines

Genie Code is your expert data engineer, built to help you design and evolve reliable data pipelines.

- Create pipelines from natural language: Describe what you need and Genie Code generates a complete Spark Declarative Pipeline with ingestion, transformations, and data quality expectations built in.
- Extend existing pipelines: Add datasets, modify transformations, write AutoCDC flows, configure Auto Loader, and apply data quality expectations, all within the context of your current pipeline.
- Understand pipeline behavior: Inspect outputs, trace data flow into downstream tables, and surface unexpected changes in row counts or schemas.

The screenshot displays the Databricks Genie Code interface. On the left, a file explorer shows a pipeline named 'New Pipeline 2026-...' with transformations organized into bronze, gold, and silver layers. The main editor shows a SQL query for 'gold_property_performance' (quality = 'gold') that aggregates booking counts, revenue, and review ratings by property ID. On the right, a summary of 18 actions is provided, detailing the Bronze Layer (ingesting raw data) and the Silver Layer (cleaned and enriched data). A chat window at the bottom right shows a conversation about creating silver properties and running the pipeline.

```

1 gold_property_performance
2 TBLPROPERTIES (quality = 'gold')
3 AS
4 SELECT
5   b.property_id,
6   COUNT(b.booking_id) AS
7   booking_count,
8   SUM(b.total_amount) AS
9   total_revenue,
10  AVG(r.rating) AS
11  average_review_rating,
12  AVG(b.base_price) AS
13  average_base_price
14 FROM silver_bookings b
15 LEFT JOIN bronze_reviews r
16   ON b.property_id = r.property_id
17 GROUP BY b.property_id;

```

property_type, destination_id, base_price), with an expectation that check_out > check_in. Create silver_properties joining properties with destinations for country and state.
Gold: create gold_property_performance aggregating per property the booking count, total revenue, average review rating, and average base_price from the silver tables.
Then, run the full pipeline

18 actions
Successfully created and executed your medallion architecture pipeline! Here's what was built:

Bronze Layer (ingests from `samples.wanderbricks`):

- `bronze_properties` - raw property data
- `bronze_bookings` - raw booking data
- `bronze_reviews` - raw review data

Silver Layer (cleaned and enriched):

- `silver_bookings` - bookings joined with properties to add `property_type`, `destination_id`, and `base_price`. Includes a data quality expectation

@ for objects, / for commands, ↑↓ for history

Agent ▾ ▶

Always review the accuracy of responses.

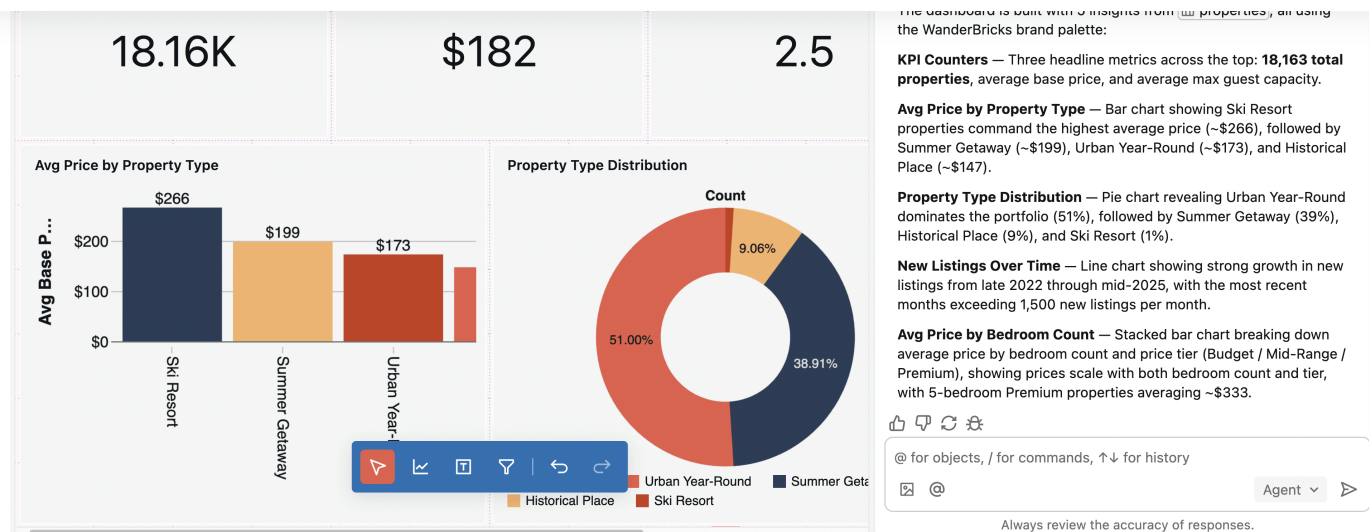


Genie Code has moved us beyond assisted coding into true agentic data engineering. It can analyze our Lakeflow pipelines, propose multi-file changes with diffs, execute runs with safeguards, and iterate through failures until issues are resolved. It feels less like autocomplete and more like a collaborator embedded in our workflow.

— Nishit Gajjar, Tech Lead, Global Infrastructure Technology Provider

Create Dashboards with Reusable Semantic Definitions

Genie Code can generate visualizations, configure filters, and organize multi-page dashboard layouts, all with reusable semantic definitions. It connects those definitions to filters, calculations, and layouts that scale as dashboards grow, helping teams move faster while maintaining consistency.

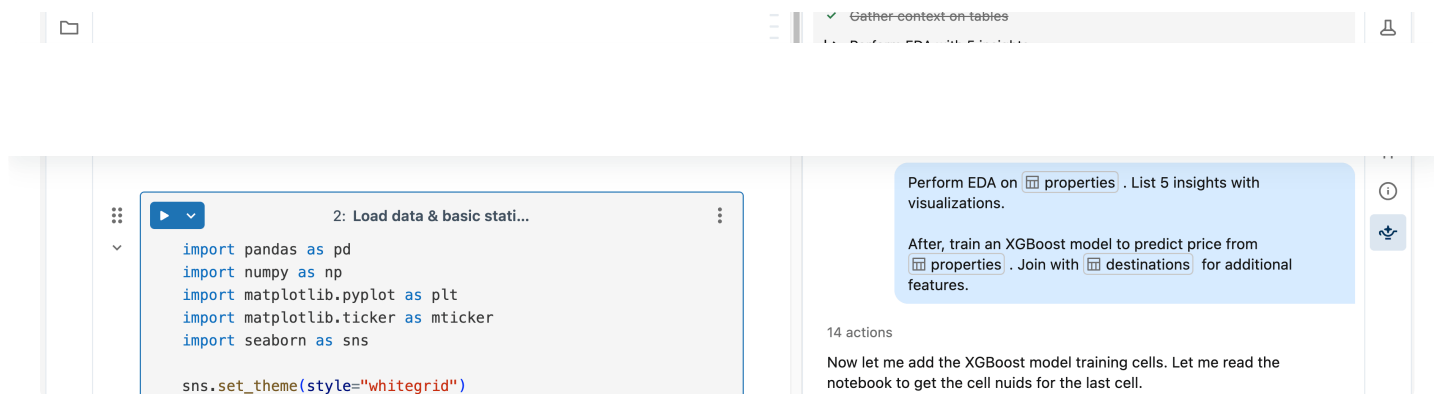


With Genie Code, our teams are delivering AI-driven analytics and automated workflows in weeks, not months. Low-code agents help us move faster while staying aligned to governance, enabling project and engineering teams to get natural-language insights from complex data without slowing delivery.

— Russell Singer, Chief Data Architect, Bechtel Corporation

Autonomous Multi-Step Planning and Execution

Provide a high-level objective, such as "Identify flight delay risks and build a monitoring dashboard". Genie Code reasons through the requirements, formulates a multi-step plan, and executes it across all Databricks Notebooks, AI/BI Dashboards, and Lakeflow in a single conversation thread.

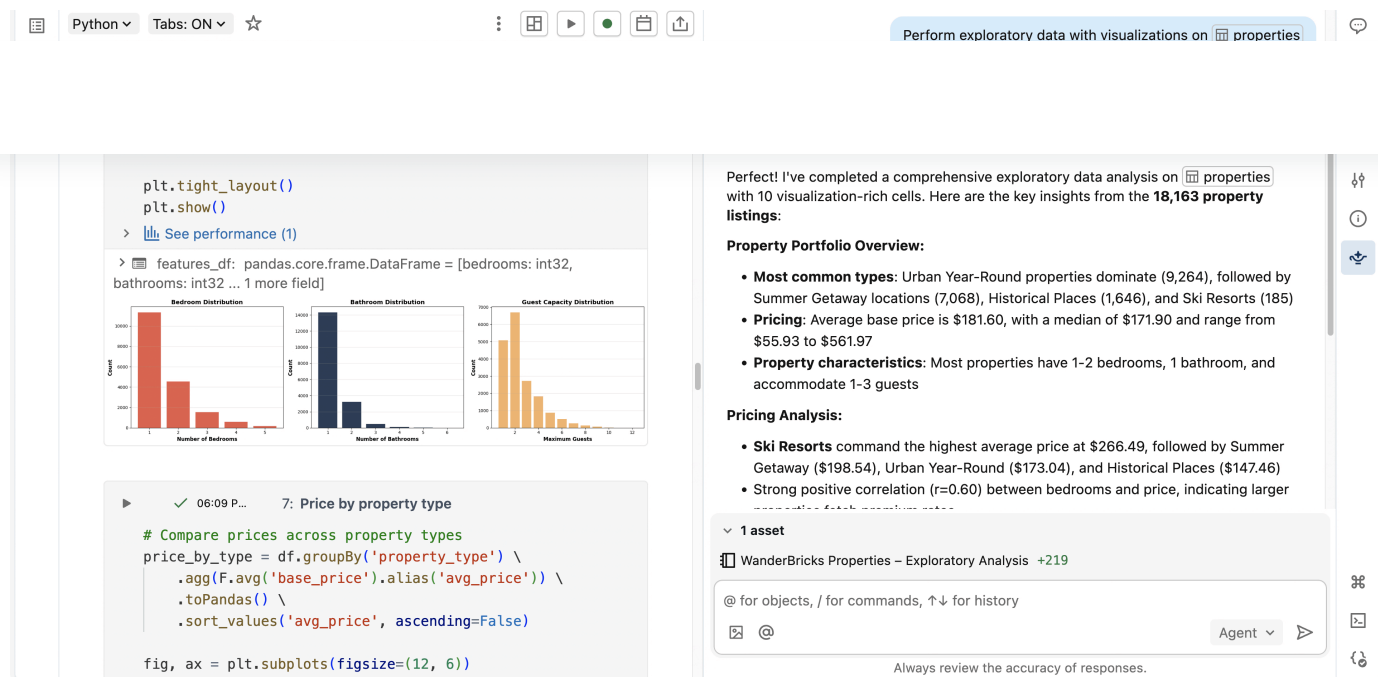


What we're seeing at Danfoss is that Genie Code changes the roles inside a data team, supporting our strategic focus on digitalization and AI. Data scientists still provide direction and review, but engineers, analysts, and domain experts can now actively work in notebooks with the assistant and contribute to advanced analytics workflows. It turns data science into a much more collaborative team activity.

— Radu Dragusin, Principal Engineer, Data & AI, Danfoss

Exploratory Data Analysis with Deep Contextual Search

Genie Code uses popularity, lineage, code samples, and Unity Catalog metadata to find the most relevant datasets for any analysis. This deep contextual search eliminates the manual effort of hunting for data and ensures that your work is based on the most accurate and frequently used tables within your organization.



”

I’m genuinely mesmerized. Genie Code feels like a glimpse into the future of how data work gets done.

— Sameer Yasser, Sr. Data Engineer, Sundt Construction

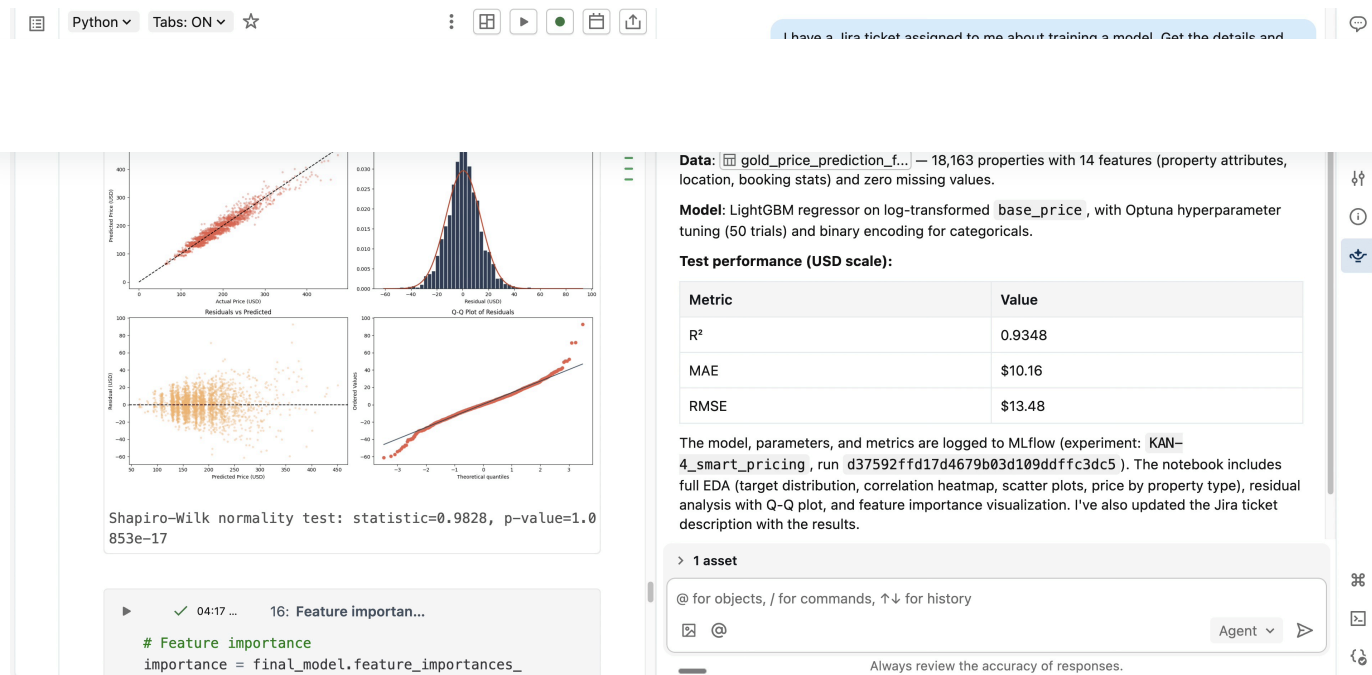
Customization and Extensibility

Genie Code is a flexible platform designed to be tailored to your team’s specific standards and external tech stack. There are three primary ways to extend its capabilities:

1. External Tooling via Model Context Protocol (MCP)

Genie Code supports [Model Context Protocol \(MCP\)](#), an open standard that allows it to securely interact with your external tools, APIs, and documentation. This enables autonomous workflows that extend beyond the Databricks workspace.

For example, when you’re assigned a Jira task to train a new ML model, Genie Code can automatically gather context from it, perform the task, and update the ticket with the results.



Connect Genie to your internal Confluence, Google Drive, GitHub, or Notion via MCP so it can reference your team's specific runbooks and data dictionaries when troubleshooting.

- 2. Agent Skills:** Define [domain-specific capabilities](#) to teach Genie Code how to perform complex tasks consistently. Whether it's a specific way your company handles PII masking or a custom framework for data validation, Skills ensure that the AI follows your organization's best practices every time. Skills follow the open Agent Skills [format](#).
- 3. Memory:** Genie Code grows smarter the more you use it. Through persistent memory, the agent automatically updates its internal instructions based on your past interactions. It learns your coding preferences, remembers which datasets you use most frequently, and retains context across sessions.

From Code to Production: Observability with Genie Code

Writing code is only the first step. Maintaining it is the real challenge. Genie Code acts as an observability agent to keep your data and AI workflows healthy. While thousands of customers use Databricks to serve sophisticated AI applications, debugging those models in production is often the most time-consuming part of the lifecycle.

Genie Code now integrates directly with Databricks Model Serving and MLflow 3.0 to automate this process. Instead of manually searching through logs and traces, you can

Endpoint health check: Get a full status report across compute, request handling and

Serving endpoints >

test-xgboost

⋮

Alerts

Permissions

Edit

Stop

Use

▼

✓ Ready

<https://ai-bi-demo-default.cloud.databricks.com/serving-endpoints/test-xgboost/invocati...>

Gateway

Edit AI Gateway

Usage monitoring:

system.serving.endpoint_usage

Rate limits:

Not enabled

Dimension table:

system.serving.served_entities

Inference tables:

Not enabled

Active configuration

Entity	Version	Name	State	Compute	Traffic (%)
pm_demo_default.default.xgboost-test	Version 1	xgboost-test-1	✓ Ready	CPU, Custom 4 concurrency	100

Metrics Events Logs

Genie Code

+ ⚙ ⋮ ✕

Perform a health check on this endpoint and identify any issues

Thoughts: The user wants a health check on the `test-xgb...

I'll run a comprehensive health check by gathering health status, configuration details, recent metrics, and AI Gateway config in parallel.

4 actions

Health Check Summary: test-xgboost

✓ Healthy

Your endpoint is **READY** and was successfully deployed just ~30 minutes ago (today at 3:58 PM PST). The deployment completed cleanly in about 9 minutes with no errors. CPU and memory usage are very low (~0.9% CPU, ~4.5% memory), and no request traffic or errors have been recorded in the last 24 hours — likely because the endpoint is brand new.

⚠ Recommendations for Improvement

1. Increase Provisioned Concurrency (High Priority)

Your min and max concurrency are both set to **4**. For production workloads, a minimum of **8 provisioned concurrency** is recommended for reliability and performance. Also, set a higher max concurrency to match

@ for objects, / for commands, ↑↓ for history

📎 @

Agent ▼

➤

Always review the accuracy of responses.

- **Agent quality analysis:** Surface subtle issues like hallucinations, incorrect tool calls, and user frustration patterns across complex agent traces in real time.

← prod_telco_support_agent GenAI apps & agents ⓘ Share

Training runs

Observability

Traces

Sessions

Evaluation

Judges

Datasets

Evaluation runs

Labeling schemas

Labeling sessions

Prompts & versions

Prompts

Agent versions

Logs Charts Beta Time Range: Custom ⓘ Start: 12/02/2025, 06:42 PM End: Search traces... Filters Sort: Request time Columns Actions Group by session 116 of 116

Assessments (4/7)

request ti...	State	brand_compli...	data_privacy	query_resolut...	response_clar...
	PASS	PASS	PASS		STRING
	38%	90%	41%		5 values
	Pass	38%	Pass	90%	Pass
	Fail	62%	Fail	10%	Fail
	Error	87	Error	87	Error
04/2025...	OK	Fail	Pass	Pass	excellent
04/2025...	OK	Fail	Pass	Pass	excellent
04/2025...	OK	Fail	Pass	Fail	adequate
04/2025...	OK	Fail	Pass	Fail	adequate
03/2025...	OK	Pass	Fail	Pass	excellent
03/2025...	OK	Fail	Pass	Pass	good

Genie Code

+ ⚙ ⋮ ✕

> Agent plan Assess scorer effectiveness (1 of 4)

2 actions

Great! Now I can see the issues. Let me analyze a few more traces to confirm patterns:

3 actions

Perfect! Now let me create a comprehensive analysis. I'll write a script to systematically analyze assessment patterns across all traces:

Thoughts: Let me create a summary based on what I've found and then provi...

Now let me create a comprehensive analysis script to examine all assessment patterns:

[Summary]

I've identified **three critical issues** in your telco support agent experiment:

✓ The Good News:

- **0% error rate** - all 100 recent traces completed successfully
- **tool_accuracy scorer is working** and catching real issues
- Latency is reasonable (3-8 seconds per trace)

⚠ The Issues:

1. 🚨 CRITICAL: Broken Span Instrumentation

- All spans show type "UNKNOWN" instead of proper types (LLM, TOOL, RETRIEVER, AGENT)
- Span-level durations are missing (showing "N/A")
- This prevents proper latency bottleneck analysis

2. ⚠ MODERATE: Evaluation Success Failure

@ for objects, / for commands, ↑↓ for history

📎 @

Agent ▼

➤

Always review the accuracy of responses.

Background Agents that Keep Workloads Healthy

Genie Code is designed to work in the background so that your data remains healthy even after you close your laptop. You can deploy multiple agents in parallel to handle the operational work that typically consumes a data engineer's week. These background agents move beyond reactive support toward proactive maintenance by handling repetitive tasks such as responding to job failures and managing routine upgrades. When a pipeline breaks, the agent identifies the root cause and suggests a fix only after validating it in a secure sandbox environment.

For example, if a production pipeline fails due to a schema mismatch, such as a column changing from an INT (150) to STRING ("150 USD"), Genie Code will pinpoint the failure and automatically fix the broken pipeline.

Background agents are coming soon.

Grounded in Unity Catalog: Integrated Security and Governance

Genie Code is built directly on Unity Catalog. This integration ensures that the agent follows the same security and governance rules as the rest of the Databricks platform.

When Genie Code searches for data, it only surfaces assets the user is authorized to access. When it builds a pipeline, it adheres to existing lineage and access controls.

- **Native Revision History:** Every edit is tracked through the Databricks versioning system. You can roll back changes across notebooks, queries, files, and Lakeflow pipelines with full confidence.
- **Built-in Guardrails:** Genie Code is designed to proactively ask for confirmation before executing code that can modify underlying tables.

Available in your Workspace today

Genie Code is **Generally Available** in your Databricks workspace right now. You can find the Genie Code panel in your notebooks, SQL editor, and Lakeflow Pipelines editor today —no complex configuration required.

Learn More

If you would like to learn more about Genie Code:

- Visit our [web page](#) to understand key Genie Code features and use cases and learn how it works across the Databricks platform
- Watch the [demo](#) to see Genie Code plan and execute real data workflows end to end
- Read the [documentation](#) to start using Genie Code in your own workspace today

We're excited to see what you build with Genie Code and how autonomous agents will reshape the way your data teams work in Databricks.

Get the latest posts in your inbox

Subscribe to our blog and get the latest posts delivered to your inbox.

Work Email *

Country *

Subscribe



Databricks processing my personal data in accordance with its privacy policy.

[View all blogs](#)



[Why Databricks](#) ✓

[Product](#) ✓

[Solutions](#) ✓

[Resources](#) ✓

[About](#) ✓



© Databricks 2026. All rights reserved. Apache, Apache Spark, Spark, the Spark Logo, Apache Iceberg, Iceberg, and the Apache Iceberg logo are trademarks of the Apache Software Foundation.

[Privacy Notice](#) | [Terms of Use](#) | [Modern Slavery Statement](#) | [California Privacy](#)
| [Your Privacy Choices](#) 